CRT questions

1. a) List four legal identifier names

   totalScore
   Player_name
   itemCount3
   Height

   b) List four illegal identifier names and explain why each is illegal
   2ndPlayer, Illegal because Java identifiers cannot start with a digit.
   class, Illegal because class is a reserved keyword in Java.
   total cost , illegal because spaces are not permitted in variable names.
   my-variable , Illegal because hyphens are not allowed (Java interprets - as a subtraction operator).

2. a) In two statements, declare a variable named numBeads and assign it the value 5:

```
int numBeads;
numBeads = 5;
```

   b) In one statement, declare a variable named numBeads and assign it the value 5:

```
int numBeads = 5;
```

3. a) What is the final value of yourNumber after the last statement executes?

   Answer: 13
   Explanation: myNumber starts at 5 and yourNumber starts at 4. Line 3 recalculates myNumber as 4 * 2 = 8. Line 4 updates yourNumber as 8 + 5, giving a final value of 13.

   b) What is the final value of yourNumber after the last statement executes?
   Answer: 11
   Explanation: yourNumber is initialized to 4. Line 3 calculates myNumber as 4 + 7 = 11. Line 4 assigns myNumber to yourNumber, so yourNumber becomes 11.

4. a) The number of basketballs in a department store: int (Used for whole counts)

   b) The price of a basketball: double (Used for currency values with decimals)
   c) The number of players on a basketball team: int (Used for whole counts)
   d) The average age of the players on a basketball team: double (Averages usually result in decimal numbers)

e) Whether a basketball player has received a jersey or not: boolean (Represents a true/false condition)
f) The first initial of a basketball player's first name: char (Holds a single character)

5. a) What is the difference between a primitive data type and an abstract data type?

A primitive data type is built directly into Java (like int, double, char, or boolean) and stores a single simple value in memory. An abstract data type (ADT) is a data structure or object (like a String or custom class) defined by programmers that combines data along with operations (methods) to manipulate that data.

b) What is the difference between a class and an object?
A class is the blueprint or template that defines what properties and behaviors something will have. An object is a specific instance created from that class blueprint when the program is running.

11. Type Casting and Rounding

Using the declarations:

- `int j = 5;`
- `double k = 1.6;`
- `int y;`
- `doube z;`

- **a) `y = j * k;`**
  - **Answer:** `y = (int)(j * k);`
  - **Why:** Multiplying an `int` (`j`) by a `double` (`k`) results in a `double` value. Since the target variable `y` is an `int`, an explicit cast `(int)` is required to prevent a compilation type mismatch error due to potential loss of precision.
- **b) `z = j * k;`**
  - **Answer:** `z = j * k;` *(No casting necessary)*
  - **Why:** Type casting is not needed here because `j` is automatically promoted to a `double` during the multiplication, and the resulting `double` value can be safely assigned directly to the `double` variable `z`.
- **c) `z = k * k;`**
  - **Answer:** `z = k * k;` *(No casting necessary)*

    - **Why:** Both operands are already `doubles`, so multiplying them results in a `double`, which matches the data type of z perfectly without any conversion.
- **d) `j = k;`**
    - **Answer:** `j = (int)k;`
    - **Why:** Assigning a `double` value (`k`) into an `int` variable (`j`) causes a narrowing conversion where the decimal portion is truncated, so Java strictly requires an explicit cast.
- **e) `k = j;`**
    - **Answer:** `k = j;` *(No casting necessary)*
    - **Why:** Going from an `int` to a `double` is a widening conversion that happens automatically behind the scenes because a `double` has enough capacity to hold an integer value.
- **f) `y = j + 3;`**
    - **Answer:** `y = j + 3;` *(No casting necessary)*
    - **Why:** Both `j` and the literal 3 are integers, meaning the mathematical operation evaluates strictly to an `int`, which matches the type of y with no conversion needed.